

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000825	Version:	6.0, CURRENT
	Status:	Effective	Effective Date:	12-Jul-2011
Title:	SM DG57. PCAs replacement.			Page 2 of 11

DG-57 Service Manual

PCA's REPLACEMENT

Procedure description
This procedure describes how to replace the PCB Robot XYZZ (Ver.2), PCB Centrifuge (Ver.2), PCB S&R Carrousel (Ver.2) and PCB In/Out (Ver.2). Note: Document REGD-0000825_v6 replaces document REGD-0000825_v5.

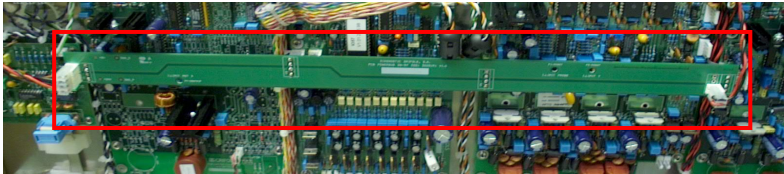
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PROCEDURE 1: PCB ROBOT XYZZ (VER.2) REPLACEMENT

EQUIPMENT

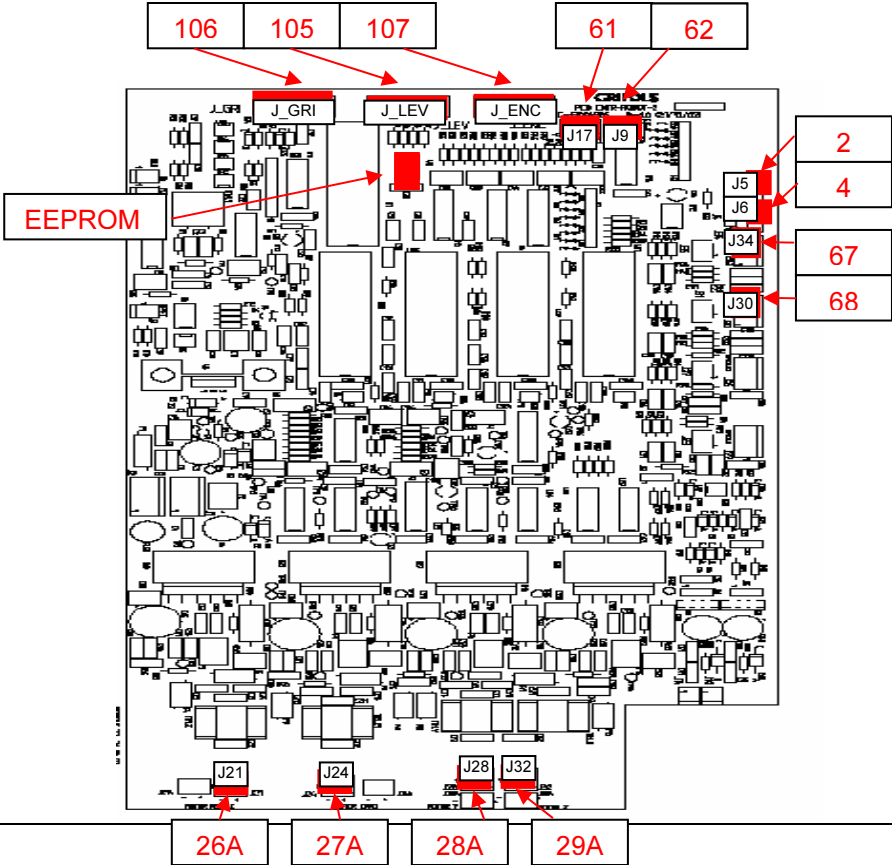
1. PCB Robot XYZZ Spare part codes:
-232225 PCB ROBOT XYZZ (Vers. 2, Firmware 1.03)
-232219 PCB ROBOT XYZZ (Vers. 2, Firmware 3.0)

SET UP

1. Turn instrument off
2. Open the rear panel
3. Remove PCB Power Bus
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Picture 1

PROCEDURE

Process	Expected Results
1. Unplug the board connectors. In order to remember their correct position we suggest to mark them. 	

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Process	Expected Results
2. Replace the board.	---
3. The board has an EEPROM where all the adjustments are stored. If the board is replaced by a new one, all the adjustments are lost. Replace the EEPROM of the new board with the one from the old board.	---
4. In case EEPROM is damaged, all adjustment related to PCB Robot must be performed. See docs number REGD-0000807 and REGD-0000808	
5. Plug each connector in its correct place. If you have marked them it will be faster and easier.	---
6. Connect the Power Bus giving power supply to the boards.	---
7. Check the board with the document REGD-0000805	---

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PROCEDURE 2: PCB S&R CARROUSEL (VER.2) REPLACEMENT

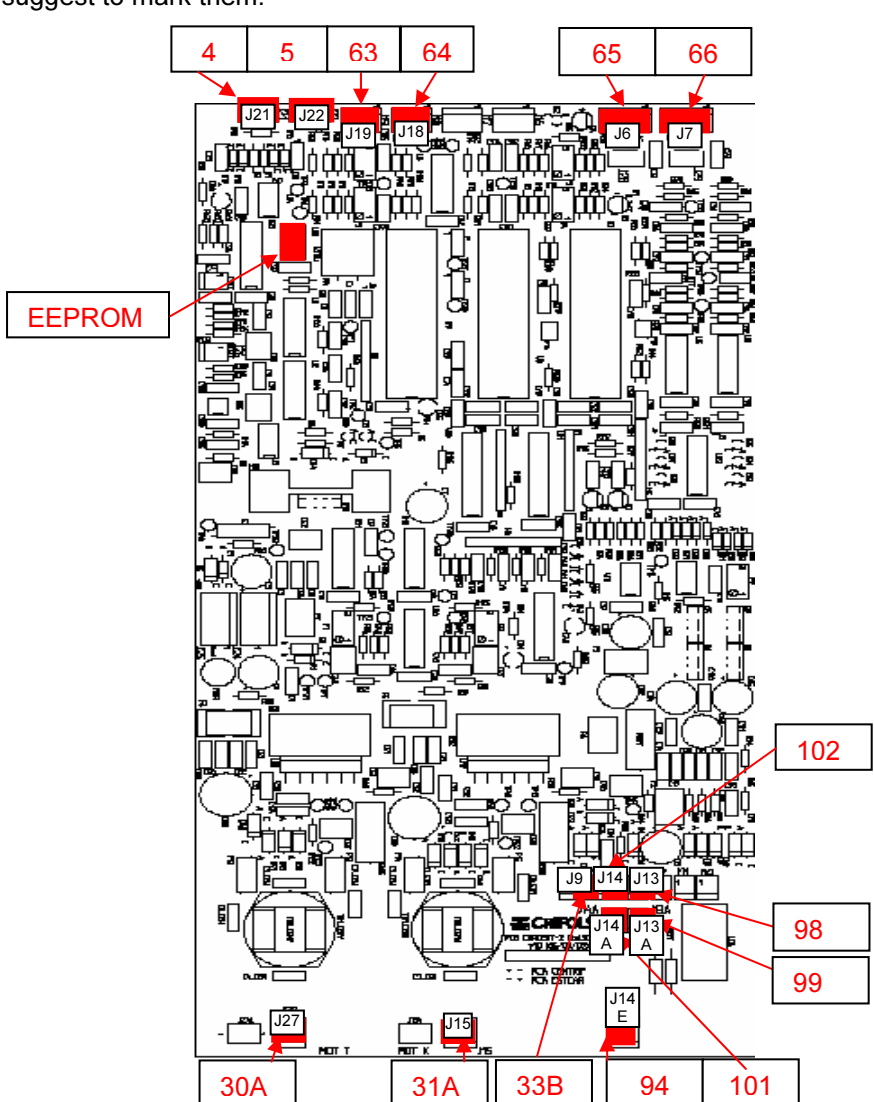
EQUIPMENT

1. PCB S&R Carrousel (Ver.2) Spare part code: 232072
2. Philips screwdriver

SET UP

1. Turn instrument off
2. Open the rear panel
3. Remove PCA Power Bus (See Picture 1)

PROCEDURE

Process	Expected Results
1. Unplug the board connectors. In order to remember their correct position we suggest to mark them. 	---

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Process	Expected Results
2. Replace the board.	---
3. The board has an EEPROM where all the adjustments are stored. If the board is replaced by a new one, all the adjustments are lost. Replace the EEPROM of the new board with the one from the old board.	---
4. In case EEPROM is damaged, all adjustments related to PCB S&R Carousel must be performed. See document REGD-0000797.	---
5. Plug each connector in its correct place. If you have marked them it will be faster and easier.	---
6. Connect the Power Bus giving power supply to the boards.	---
7. Check the board with the document REGD-0000797	---

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PROCEDURE 3: PCB CENTRIFUGE (VER.2) REPLACEMENT

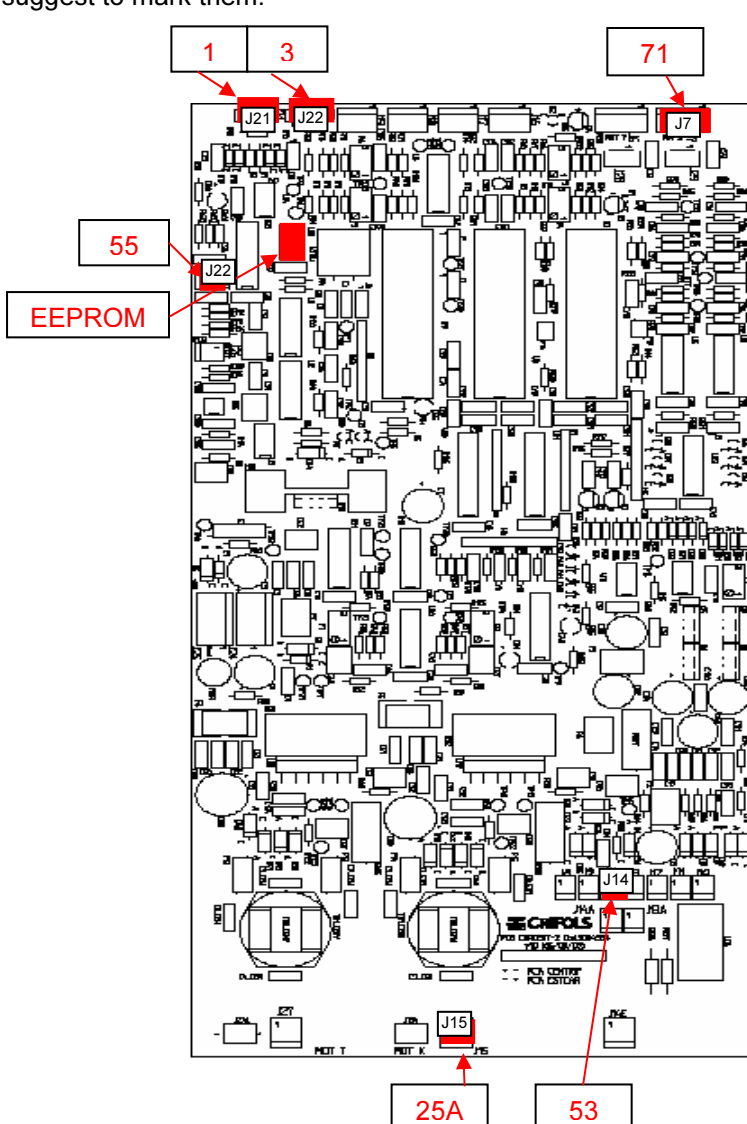
EQUIPMENT

1. PCB Centrifuge (Ver. 2, Firmware 2.0) Spare part code: 232256
2. Philips screwdriver.

SET UP

1. Turn instrument off
2. Open the rear panel
3. Remove PCA Power Bus (See Picture 1)

PROCEDURE

Process	Expected Results
1. Unplug the board connectors. In order to remember their correct position we suggest to mark them. 	---

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Process	Expected Results
2. Replace the board.	---
3. The board has an EEPROM where all the adjustments are stored. If the board is replaced by a new one, all the adjustments are lost. Replace the EEPROM of the new board with the one from the old board.	---
4. In case EEPROM is damaged, all adjustments related to PCB must be performed. See doc number REGD-0000804.	---
5. Plug each connector in its correct place. If you have marked them it will be faster and easier.	---
6. Connect the Power Bus giving power supply to the boards.	---
7. Check the board with the document REGD-0000804	---

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PROCEDURE 4: PCB IN/OUT (VER.2) REPLACEMENT

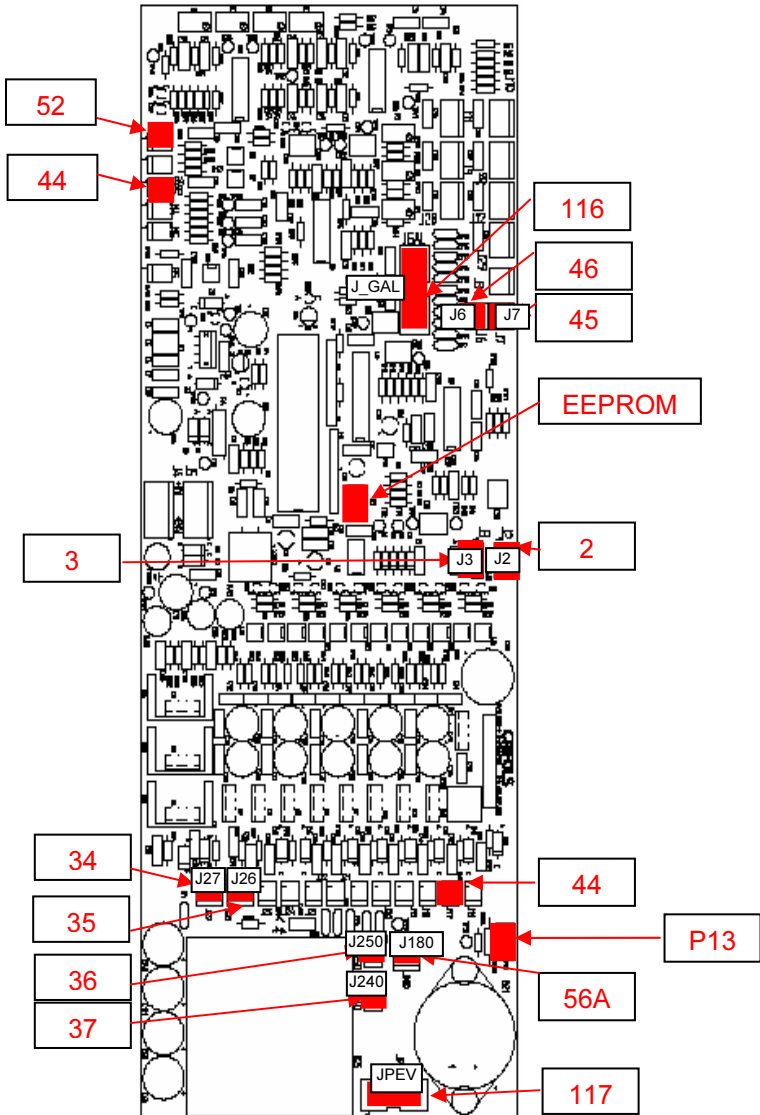
EQUIPMENT

1. PCB IN/OUT (Ver.2) Spare part code: 232073
2. Philips screwdriver.

SET UP

1. Turn instrument off
2. Open the rear panel
3. Remove PCA Power Bus (See Picture 1)

PROCEDURE

Process	Expected Results
1. Unplug the board connectors. In order to remember their correct position we suggest to mark them. 	

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Process	Expected Results
2. Replace the board.	---
3. The board has an EEPROM where all the adjustments are stored. If the board is replaced by a new one, all the adjustments are lost. Replace the EEPROM of the new board with the one from the old board.	---
4. In case EEPROM is damaged, all adjustments related to PCB must be performed. See docs number REGD-0000799, REGD-0000817, REGD-0000818, REGD-0000819 and REGD-0000821.	---
5. Plug each connector in its correct place. If you have marked them it will be faster and easier.	---
6. Connect the Power Bus giving power supply to the boards.	---
7. Check the board with the document REGD-0000799 and REGD-0000821.	---
8. Use Potentiometer P13 to adjust Buzzer volume if necessary.	---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	Added buzzer adjustment procedure in I/O PCA replacement and document reference changed.	Document REGD-0000825_v3 replaces document 57052V0M001.
4.0	2010/09/09	Changed in: - Procedure 1: Connector referenced with number 3 changed to number 2. - Procedure 3: Connector referenced with number 2 changed to number 3. - Procedure 4: Connector referenced with number 44 was in an incorrect position. Interchange connectors 2 and 3 between them.	Some references in the document were incorrect.
5.0	2011/04/28	Spare part code numbers and descriptions updated.	Some spare part codes and descriptions were obsolete.
5.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
5.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
6.0	See document header.	Document REGD-0000825_v6 replaces document REGD-0000825_v5.	Document version changed.
6.0	See document header.	Document title.	Commercial reference substituted by project reference.
6.0	See document header.	Document header.	Increase information to easy document search.